

Chapter Three

Interest rates in the Financial System

Outline

- *Types of interest rates*
 - *The theory of interest rates*
 - *The determinants of structure of interest rates*
-

Types of interest rates

- An **interest rate** is the price paid by a borrower (or debtor) to a lender (or creditor) for the use of resources during some interval.
 - what borrowers pay for the loans.
 - The amount of the loan is the **principal**, and
 - the price paid is typically expressed as a percentage of the principal per unit of time (generally, a year).
-

Types of interest rates

- ***Real rate***- the rate that would prevail in the economy if the average prices for goods and services were expected to remain constant during the loan's life.
 - The real rate is the growth in the power to consume over the life of a loan
 - ***Risk-free rate***-the rate on a loan whose borrower will not default on any obligation.
 - ***Short-term***-the rate on a loan that has one year to maturity.
 - ***Nominal Rate***-is the number of monetary units to be paid per unit borrowed, and is, in fact, the observable market rate on a loan.
-

-
- The relationship between inflation and interest rates is the well-known Fisher's Law, which can be expressed this way:

$$(1 + i) = (1 + r) \times (1 + p),$$

- Where i is nominal interest rate, r is real interest rate and p is the expected percentage change in the price level of goods and services over the loan's life
 - shows that the nominal rate, i , reflects both the real rate and expected inflation.
-

The Theory of interest rates

- Two most influential theories of the determination of the interest rate:
 - *Fisher's theory of interest*, which underlies the loanable funds theory, and
 - *Keynes's liquidity preference theory of interest*
-

Fisher's theory of interest rate

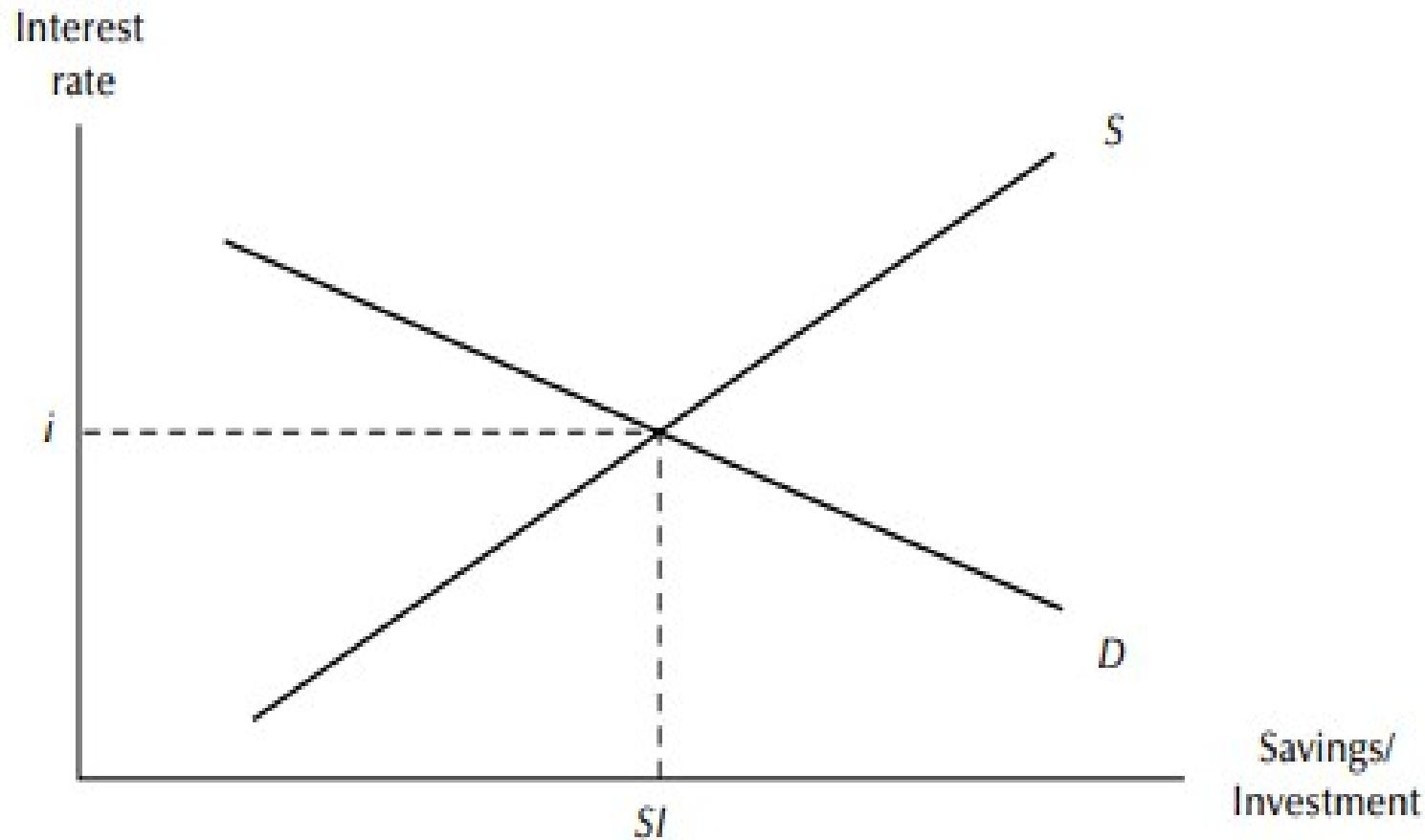
- **Saving** is the choice between current and future consumption of goods and services.
 - Individuals save some of their current income in order to be able to consume more in the future.
 - A chief influence on the saving decision is the individual's **marginal rate of time preference**
 - which is the willingness to trade some consumption now for more future consumption.
 - Individuals differ in their time preferences.
-

Fisher's theory of interest rate

- Another influence on the saving decision is **income**.
 - Generally, higher current income means the person will save more, although people with the same income may have different time preferences.
 - The third variable affecting savings is the **reward for saving**, or the **rate of interest** on loans that savers make with their unconsumed income.
 - As the interest rate rises, each person becomes willing to save more, given that person's rate of time preference.
-

Fisher's theory of interest rate

Figure 1 Equilibrium in the Market for Savings



Fisher's theory of interest rate

- There can be no reward for savings if there is no demand for borrowed resources because someone must pay the interest.
 - firms do all the borrowing, and they borrow from savers in order to invest.
 - **Investment** means directing resources to assets that will increase the firms' future capacity to produce.
-

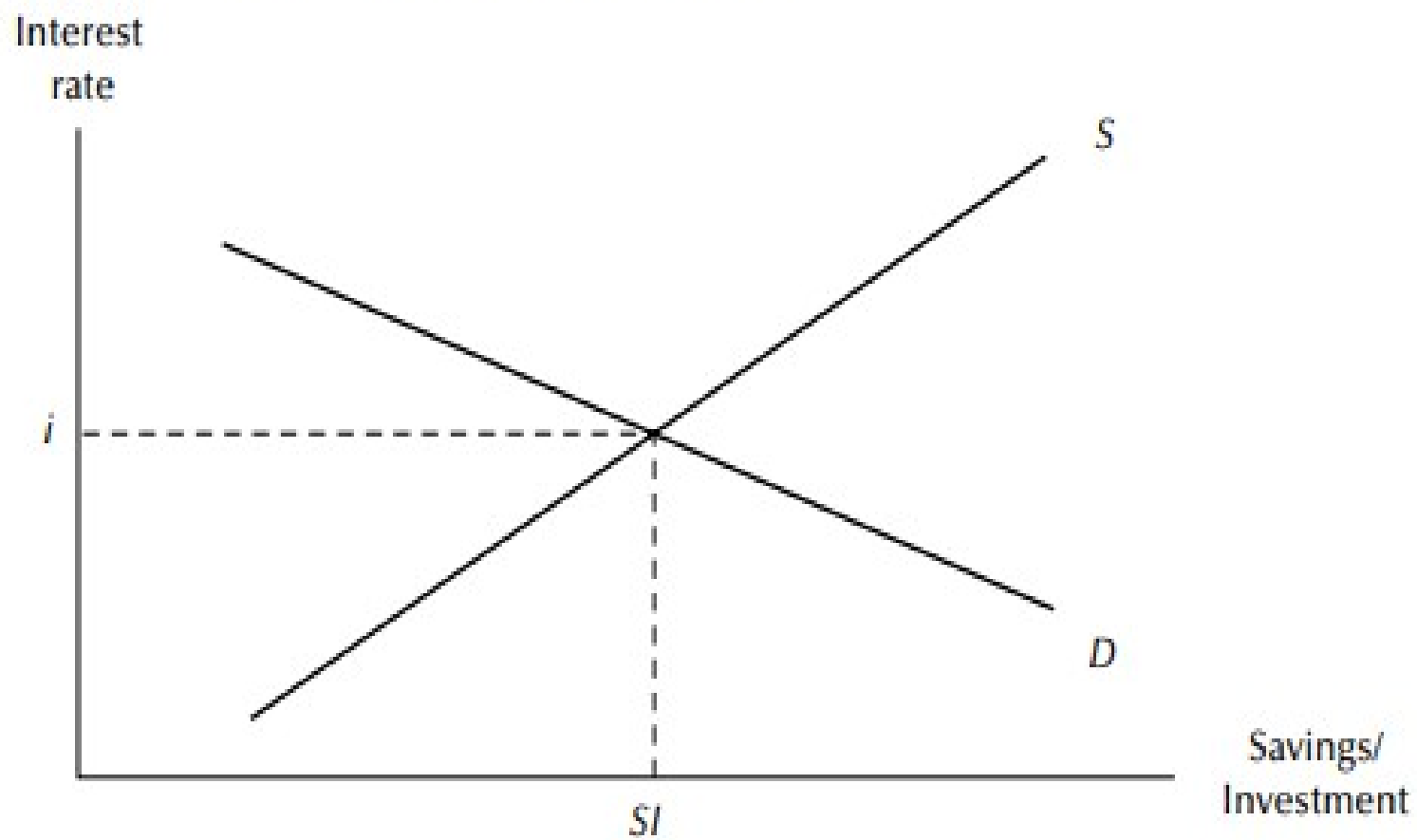
Fisher's theory of interest rate

- An important influence on the borrowing decision is the **gain from investment**, which is the positive difference between the resources used by a process and the total resources it will produce in the future.
 - The gain from additional projects, as investment increases, is the **marginal productivity of capital**, which is negatively related to the amount of investment.
 - as the amount of investment grows, additional gains necessarily fall, as more of the less profitable projects are accepted.
-

Fisher's theory of interest rate

- The maximum that a firm will invest depends on the rate of interest, which is the cost of loans.
 - The firm will invest only as long as the marginal productivity of capital exceeds or equals the rate of interest.
 - firms will accept only projects whose gain is not less than their cost of financing.
 - Thus, the firm's demand for borrowing is negatively related to the interest rate.
-

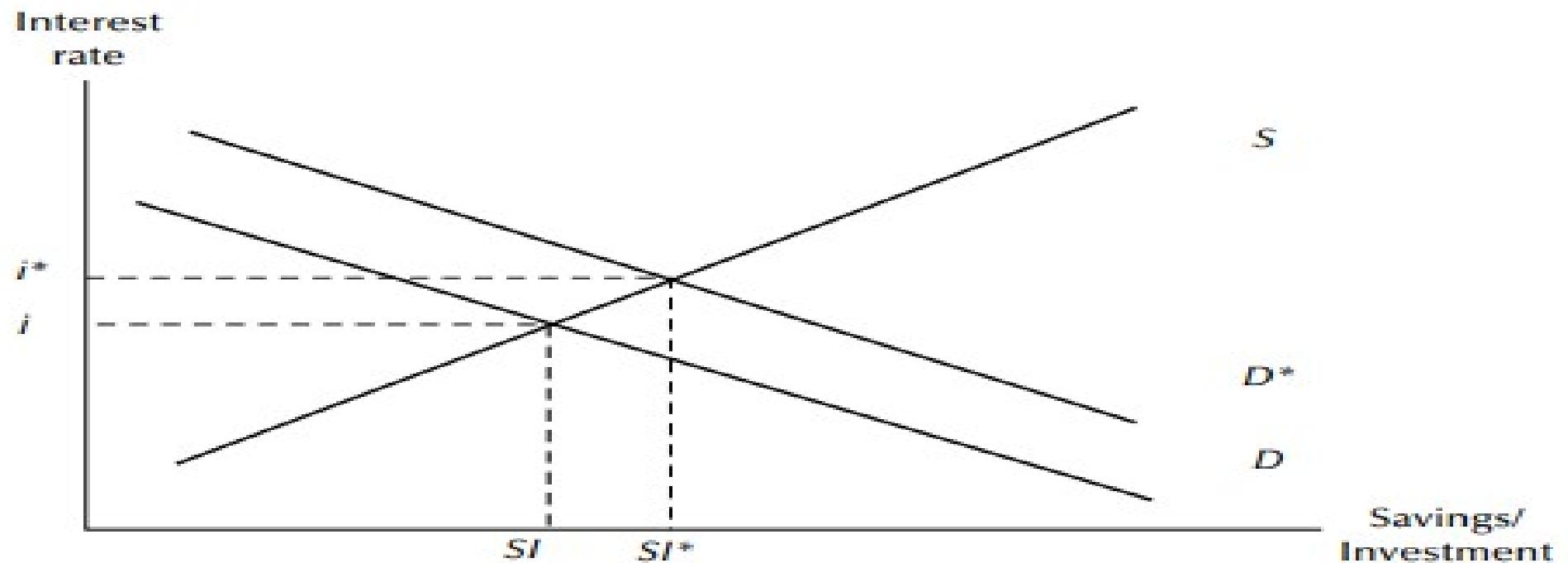
Figure 1 Equilibrium in the Market for Savings



-
- Fisher's theory emphasizes that the long-run level of the interest rate and the amount of investment depend on *a society's propensity to save* and *on technological development*.
 - increase in technological capability makes production cheaper.
 - lower production costs mean more gain on investments and a higher marginal productivity of capital.
-

- The resulting increase in firms' desired investment and borrowing through loans, at any level of the interest rate, is actually an upward shift in the demand function.

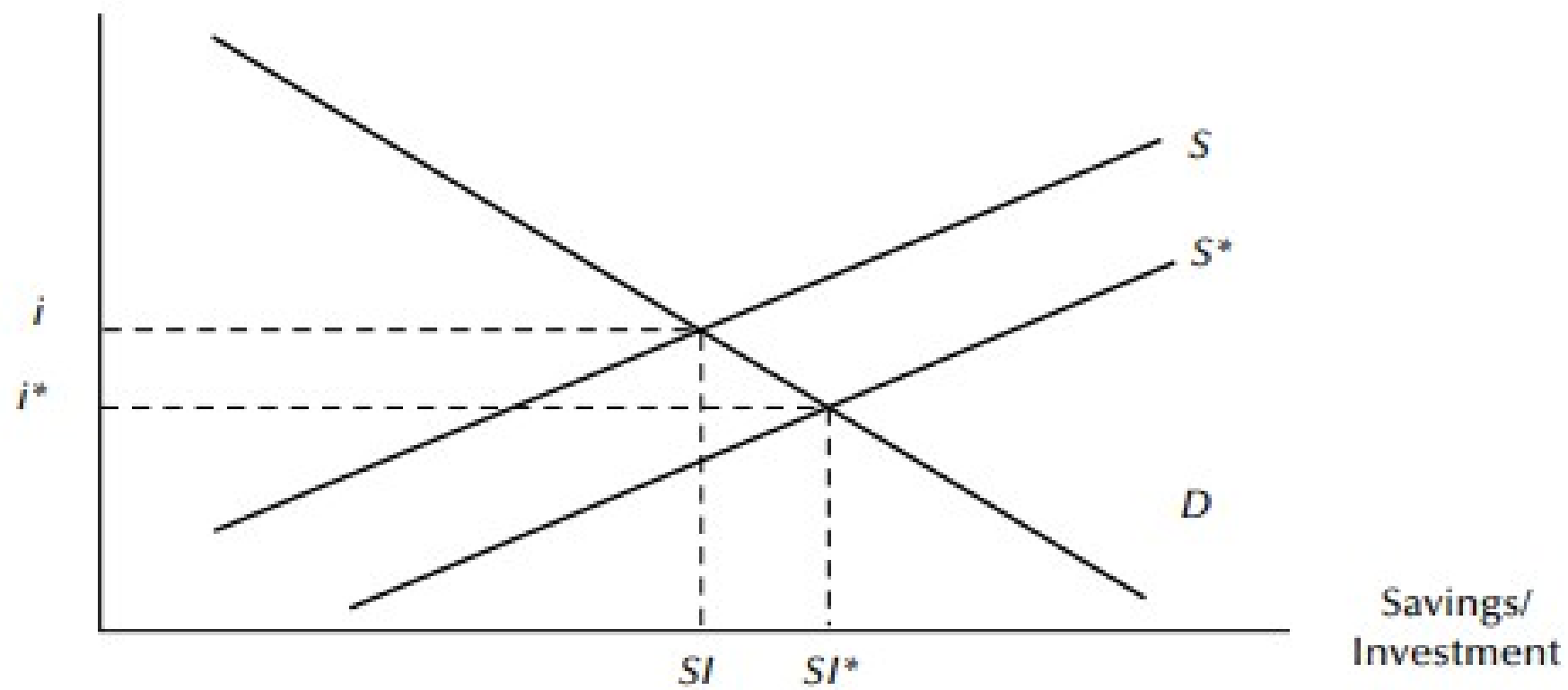
Figure 2 The Results of a Shift in the Demand for Savings



-
- When individuals suddenly become more willing to save, which amounts to a fall in the marginal rate of time preference.
 - the supply of loans function would shift downward (from S to S^*), and savings would be higher at every level of the interest rate.
 - The equilibrium interest rate would also fall, from i to i^* .
-

Figure 3 The Results of a Shift in the Supply of Savings

Interest
rate



The Loanable Funds Theory of interest rates

- Fisher's theory is a general one and obviously neglects certain practical matters, such as
 - the power of the government (in concert with depository institutions) to create money and
 - the government's often large demand for borrowed funds, which is frequently immune to the level of the interest rate.
-

The Loanable Funds Theory

- This theory proposes that the general level of interest rates is determined by the complex interaction of two forces.:
 1. the total demand for funds by firms, governments, and households (or individuals),
 - this demand is negatively related to the interest rate (except for the government's demand, which may frequently not depend on the level of the interest rate).
 2. the total supply of funds by firms, governments, banks, and individuals.
 - Supply is positively related to the level of interest rates, if all other economic factors remain the same.
-

The Loanable Funds Theory

- In an equilibrium situation:
 - the intersection of the supply and demand functions sets the interest rate level and the level of loans.
 - the demand for funds equals the supply of funds
 - This means that all agents are borrowing what they want, investing to the desired extent, and holding all the money they wish to hold.
 - In other words, equilibrium extends through the money market, the bond market, and the market for investment assets.
-

The Loanable Funds Theory

- As in **Fisher's theory**, shifts in the demand and supply curves may occur for many reasons:
 - changes in the money supply,
 - government deficits,
 - changed preferences by individuals,
 - new investment opportunities,
 - the expectation of inflation can affect the equilibrium rate through the supply of funds curve, as savers demand higher rates (because of inflation) for any level of savings.
-

The Liquidity Preference Theory

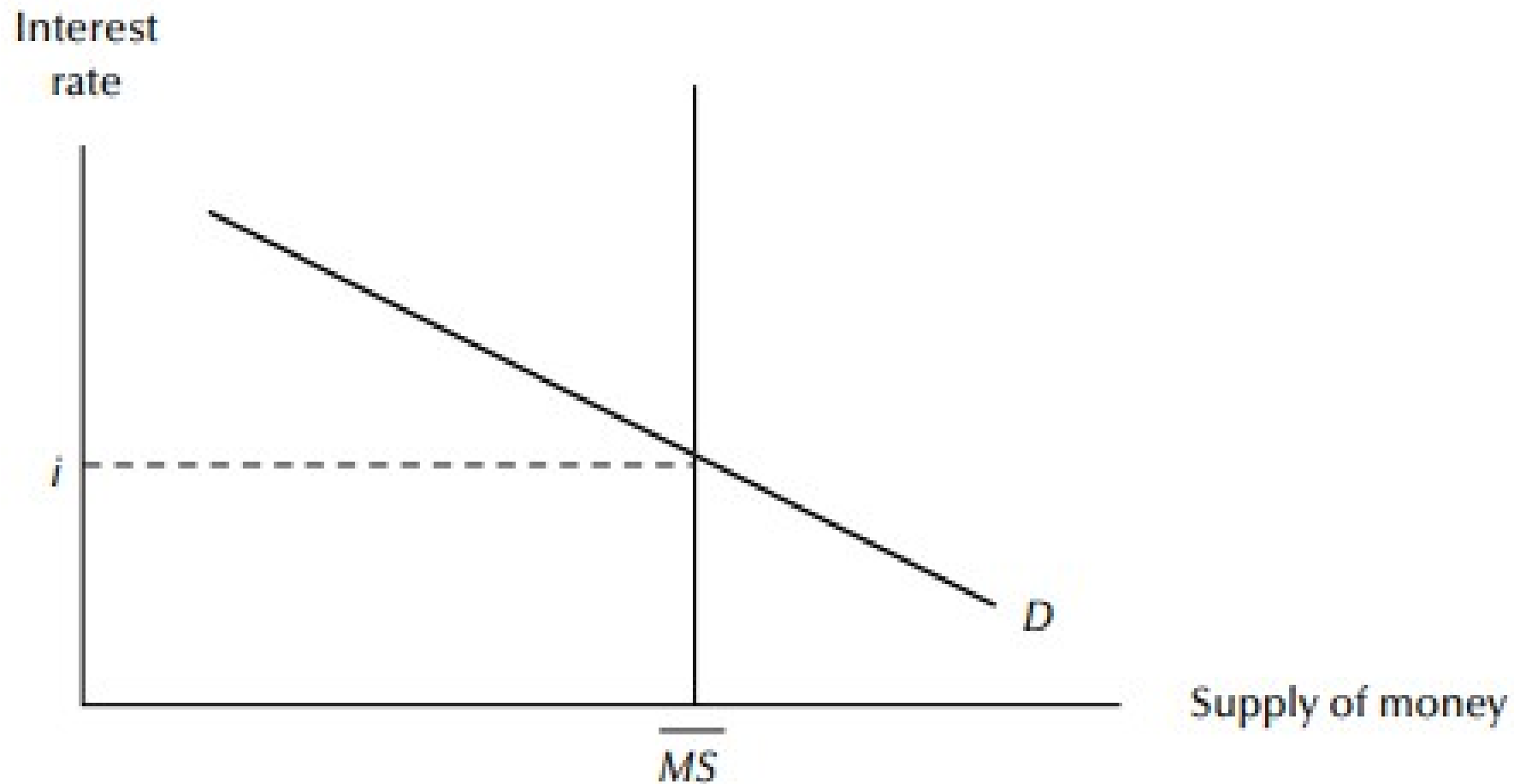
- ❑ Originally developed by John Maynard Keynes
 - ❑ analyzes the equilibrium level of the interest rate through the interaction of the supply of money and the public's aggregate demand for holding money.
 - ❑ Keynes assumed that most people hold wealth in only two forms: "money" and "bonds."
-

Demand, Supply, and Equilibrium

- The public (consisting of individuals and firms) holds money for several reasons:
 - ease of transactions,
 - precaution against unexpected events, and
 - speculation about possible rises in the interest rate.
- Although money pays no interest, the demand for money is a negative function of the interest rate.
 - At a low rate, people hold a lot of money because they do not lose much interest by doing so and because the risk of a rise in rates (and a fall in the value of bonds) may be large.
 - With a high interest rate, people desire to hold bonds rather than money, because the cost of liquidity is substantial in terms of lost interest payments and because a decline in the interest rate would lead to gains in the bonds' values.

Demand, Supply, and Equilibrium

Figure 4 Equilibrium in Keynes's Market for Money



Shifts in the Rate of Interest

- Equilibrium in the money market requires, of course, that the total demand for money equals total supply.
 - The equilibrium rate of interest can change if there is a change in any variable affecting the demand or supply curves.
 - On the demand side, Keynes recognized the importance of two such variables:
 - the level of income and the level of prices for goods and services.
 - A rise in income (with no other variable changing) raises the value of money's liquidity and shifts the demand curve to the right, increasing the equilibrium interest rate.
 - Because people want to hold amounts of “real money,” or monetary units of specific purchasing power, a change in expected inflation would also shift the demand curve to the right and raise the level of interest
-

Shifts in the Rate of Interest

- The money supply curve can shift, in Keynes's view, only by actions of the central bank.
 - The central bank's power over interest rates arises because of its ability to buy and sell securities (open market operations), which can alter the amount of money available in the economy.
 - Generally, Keynes thought that an increase in the money supply would, by shifting the supply curve to the right, bring about a decline in the equilibrium interest rate.
 - Similarly, he reasoned that a reduction in the money supply would raise rates.
-

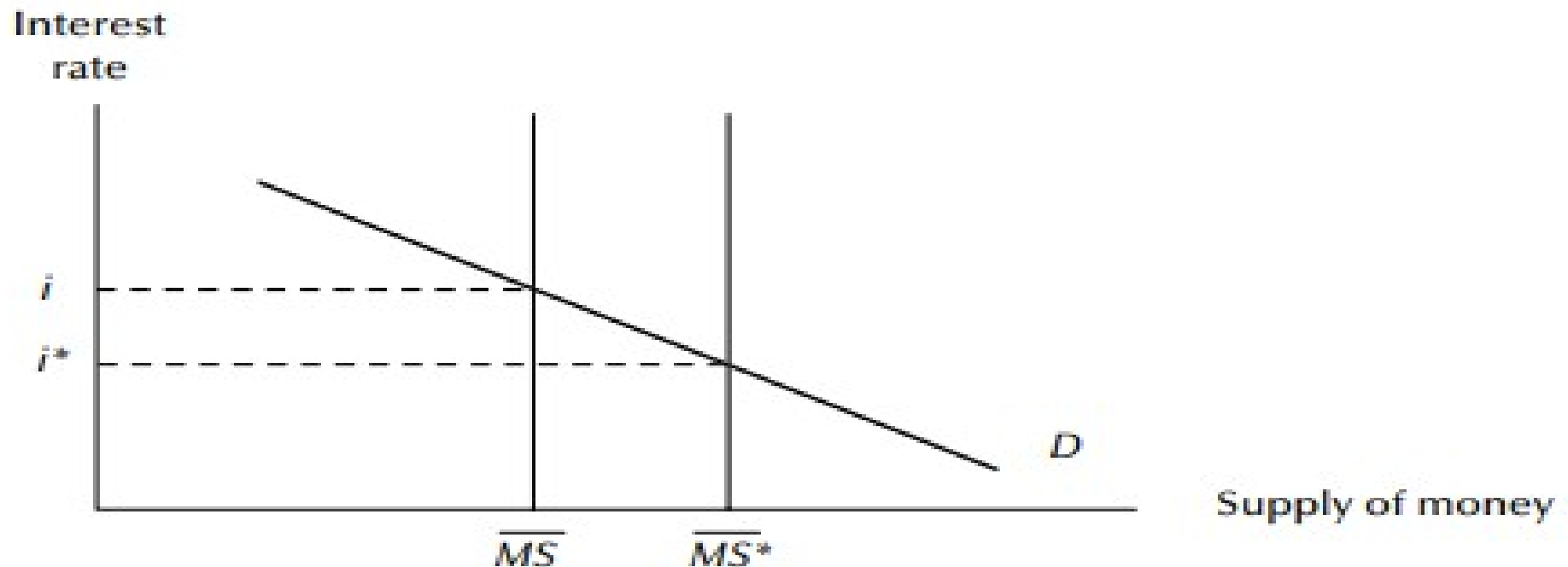
Changes in the Money Supply and Interest Rates

- A change in the money supply has three different effects upon the level of the interest rate: the liquidity effect, the income effect, and the price expectations effect.
 - These effects do not usually occur in a simultaneous manner but rather tend to be spread out over some time period following the change in the money supply.
 - These effects move rates in different ways and to different extents.
 - One effect may even cancel or overwhelm an earlier effect.
-

Liquidity Effect

- represents the initial reaction of the interest rate to a change in the money supply.
- With an increase in the money supply, the initial reaction should be a fall in the rate

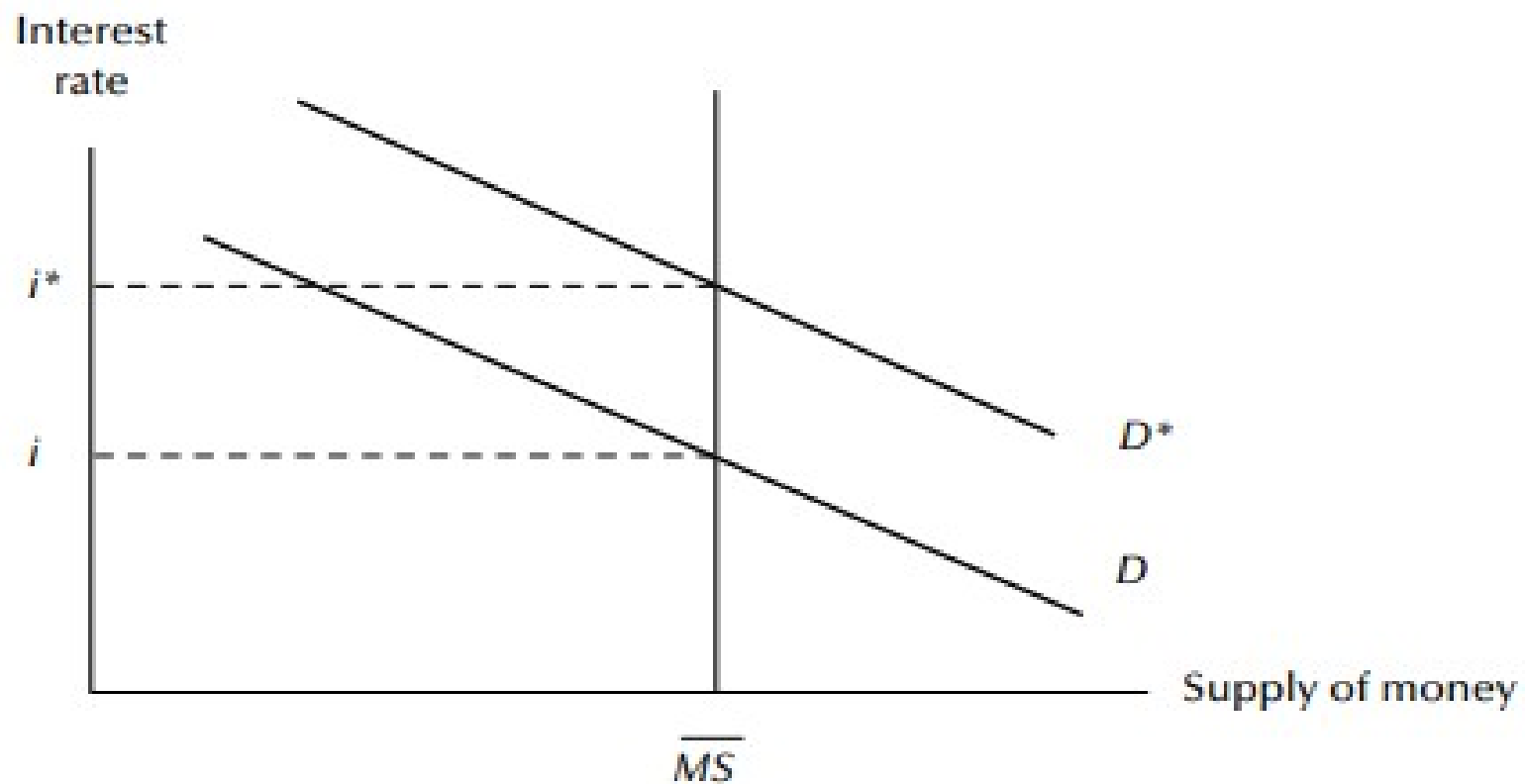
Figure 5 The Liquidity Effect of an Increase in the Money Supply



Income Effect

- It is well known that changes in the money supply affect the economy.
 - A decline in the supply would tend to cause a contraction.
 - An increase in the money supply, generally speaking, is economically expansionary:
 - More loans are available and extended, more people are hired or work longer, and consumers and producers purchase more goods and services.
 - Thus, money supply changes can cause income in the system to vary.
 - an increase in the money supply, raises income.
 - Because the demand for money is a function of income, a rise in income shifts the demand function and
.....increases the amount of money that the public will want to hold at any level of the interest rate.
-

Figure 6 The Income Effect of a Change in the Money Supply



Price Expectations Effect

- Although an increase in the money supply is an economically expansionary policy, the resultant increase in income depends substantially on the **amount of slack** in the economy at the time of the central banks (NBE's) action.
 - the price expectations effect usually occurs only if the money supply grows in a time of high output.
 - Because the price level (and expectations regarding its changes) affects the money demand function, the price expectations effect is an increase in the interest rate.
-

Risk Structure of Interest Rates

There is not one interest rate in any economy.

There, rather, is a structure of interest rates.

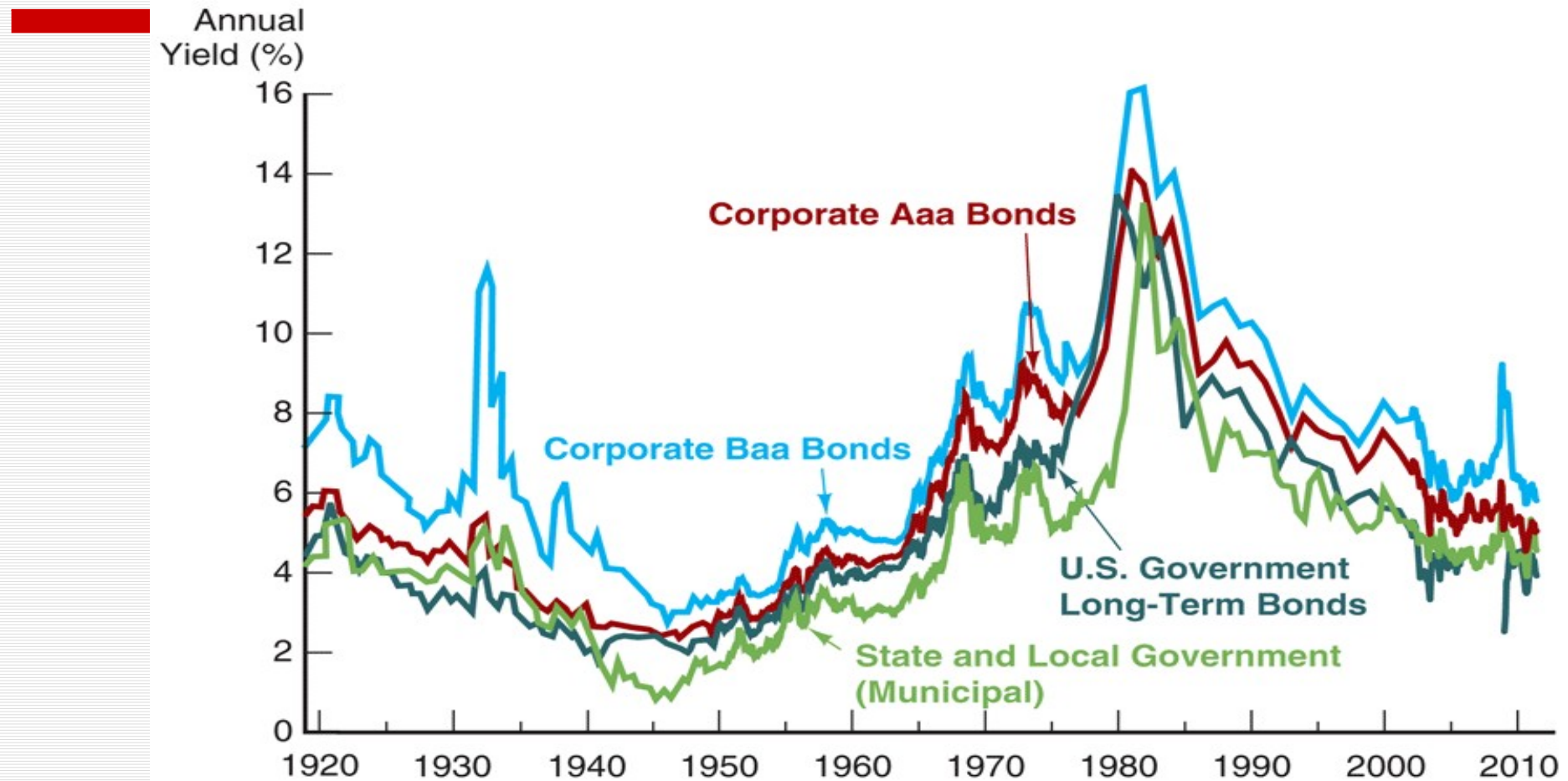
Bonds with the same maturity have different interest rates due to:

Default risk

Liquidity

Tax considerations

Figure 1 Long-Term Bond Yields, 1919–2011



Sources: Board of Governors of the Federal Reserve System, Banking and Monetary Statistics, 1941–1970; Federal Reserve; www.federalreserve.gov/releases/h15/data.htm.

Risk Structure of Interest Rates

Default risk: probability that the issuer of the bond is unable or unwilling to make interest payments or pay off the face value

U.S. Treasury bonds are considered default free (government can raise taxes).

Risk premium: the spread between the interest rates on bonds with default risk and the interest rates on (same maturity) Treasury bonds

Figure 2 Response to an Increase in Default Risk on Corporate Bonds

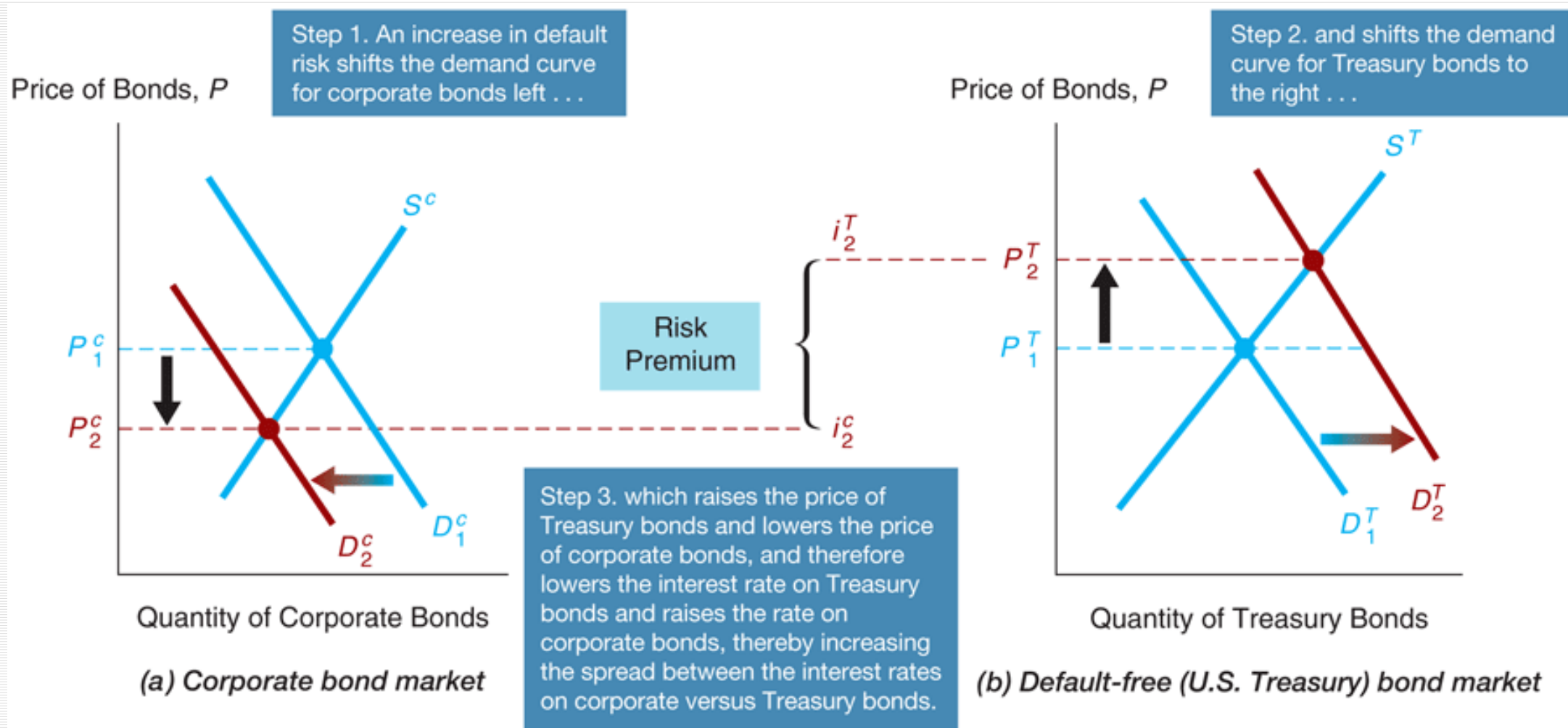


TABLE
Bond
Ratings by
Moody's,
Standard
and Poor's,
and Fitch

	Moody's	Standard & Poor's	Description
Investment Grade	Aaa	AAA	Bonds of the best quality with the smallest risk of default. Issuers are exceptionally stable and dependable.
	Aa	AA	Highest quality with slightly higher degree of long-term risk.
	A	A	High-medium quality, with many strong attributes but somewhat vulnerable to changing economic conditions.
	Baa	BBB	Medium quality, currently adequate but perhaps unreliable over the long term.
Noninvestment, Speculative Grade	Ba	BB	Some speculative element, with moderate security but not well safeguarded.
	B	B	Able to pay now but at risk of default in the future.
Highly Speculative	Caa	CCC	Poor quality, clear danger of default.
	Ca	CC	Highly speculative quality, often in default.
	C	C	Lowest-rated, poor prospects of repayment though may still be paying.
	D	D	In default.

Risk Structure of Interest Rates (cont'd)

Liquidity: the relative ease with which an asset can be converted into cash

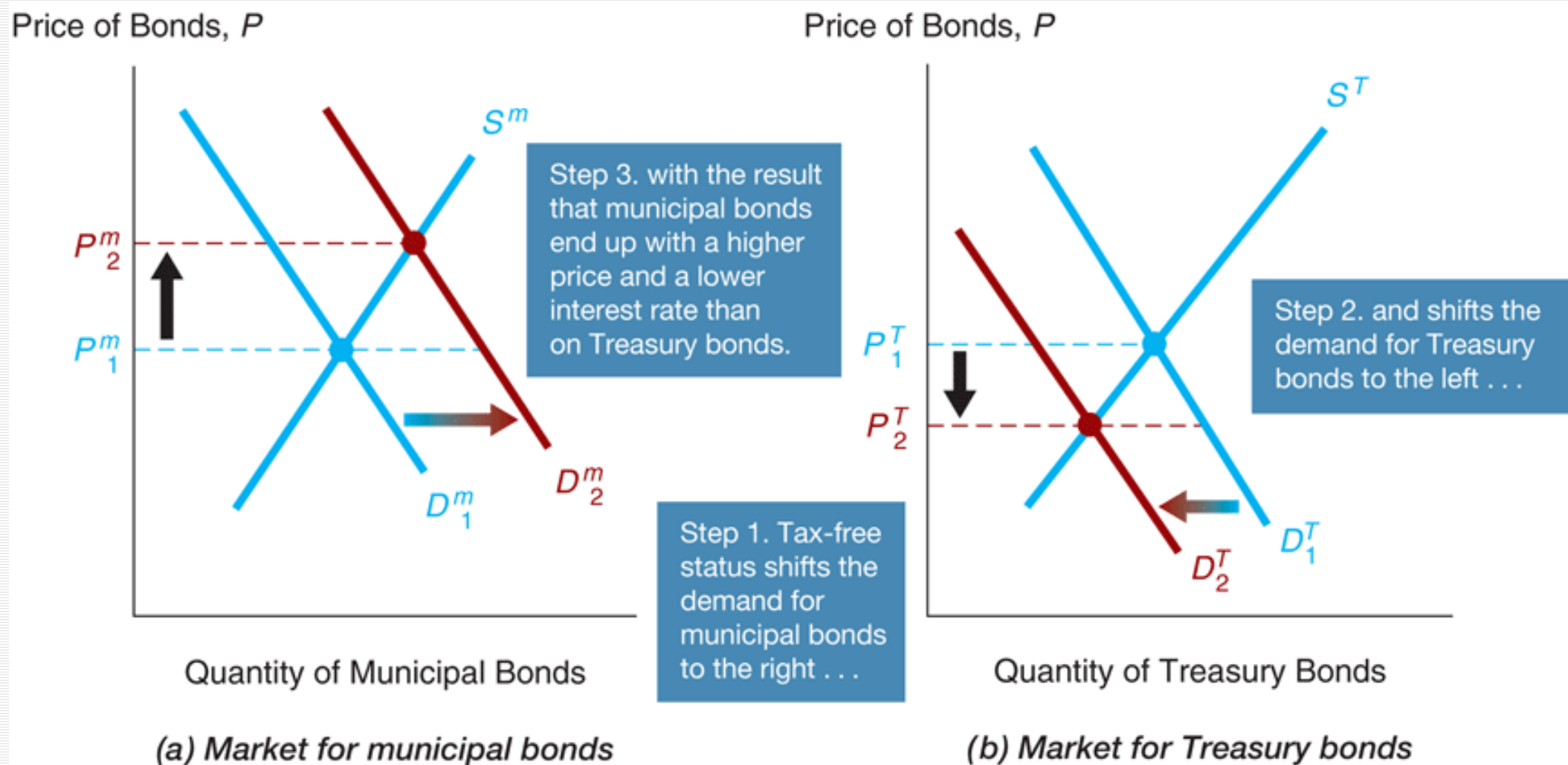
- Cost of selling a bond

- Number of buyers/sellers in a bond market

Income tax considerations

- Interest payments on municipal bonds are exempt from federal income taxes.

Figure 3 Interest Rates on Municipal and Treasury Bonds



DETERMINANTS OF THE STRUCTURE OF INTEREST RATES

- ❑ The interest rate that a borrower will have to pay will depend on a myriad of factors.
 - ❑ A bond's tax status and rating aren't the only factors that affect its yield.
-

Term Structure of Interest Rates

- Why do bonds with the same default rate and tax status but different maturity dates have different yields?
 - Long-term bonds are like a composite of a series of short-term bonds.
 - Their yield depends on what people expect to happen in the future.
-

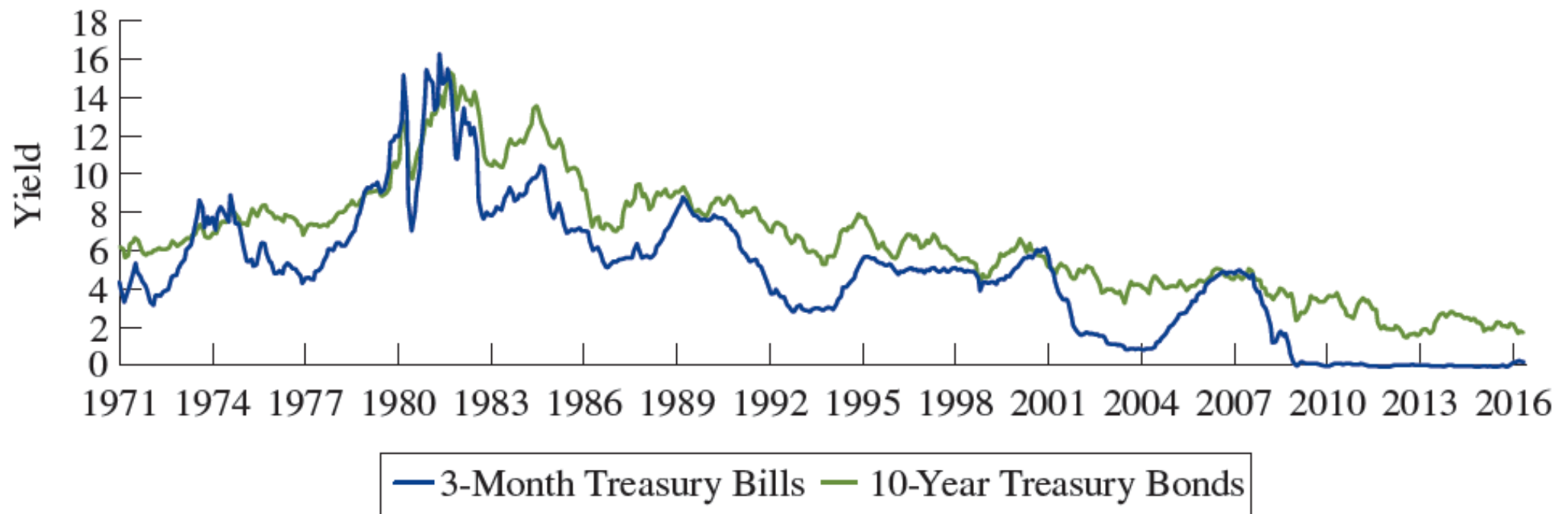
Term Structure of Interest Rates

- The relationship among bonds with the same risk characteristics but different maturities is called the **term structure of interest rates**.
- Comparing 3-month and 10-year Treasury yields we can see:
 1. Interest rates of different maturities tend to move together.
 2. Yields on short-term bonds are more volatile than yields on long-term bonds.
 3. Long-term yields tend to be higher than short-term yields.

Term Structure of Interest Rates

Figure 7.3

The Term Structure of Treasury Interest Rates



The Expectations Hypothesis

- The expectations hypothesis of the term structure focuses on the risk-free interest rate.
 - The risk-free interest rate can be computed, assuming there is not uncertainty about the future.
-

The Expectations Hypothesis

- If there is no uncertainty, then an investor will be indifferent between holding a two-year bond or a series of two one-year bonds.
 - Certainty means that the bonds of different maturities are perfect substitutes for each other.
 - The expectations hypothesis implied that the current two-year interest rate should equal the average of current one-year rate and the one-year interest rate one year in the future.
-

The Expectations Hypothesis

- When interest rates are expected to rise, long-term interest rate will be higher than short-term interest rates.
 - The **yield curve** which plots the yield to maturity on the vertical axis and the time to maturity on the horizontal axis, will slope up.
 - This also means:
 - If interest rates are expected to fall, the yield curve will slope down.
 - If expected to stay the same, the yield curve will be flat.
-

The Expectations Hypothesis

- If bonds of different maturities are perfect substitutes for each other, then we can construct investment strategies that must have the same yields.
 1. Invest in a two-year bond and hold it to maturity
 2. Invest in two one-year bonds, one today and one when the first matures.
-

The Expectations Hypothesis

- The expectations hypothesis tells us investors will be indifferent between the two options.

- This means they must have the same return:

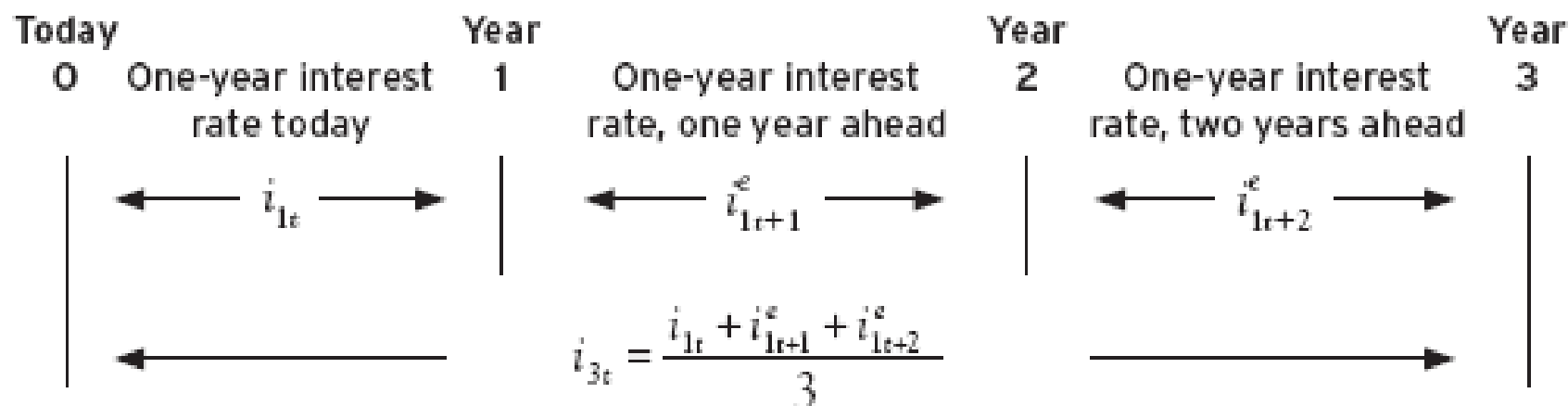
$$(1 + i_{2t})(1 + i_{2t}) = (1 + i_{1t})(1 + i_{1t+1}^e)$$

- We can now write the two-year interest rate as the average of the current and future expected one-year interest rates:

$$i_{2t} = \frac{i_{1t} + i_{1t+1}^e}{2}$$

The Expectations Hypothesis

The Expectations Hypothesis of the Term Structure:



Three-year interest rate = Average of three one-year rates

The Expectations Hypothesis

- We can generalize this: a bond with n years to maturity is the average of n expected future one-year interest rates:

$$i_{nt} = \frac{i_{1t} + i_{1t+1}^e + i_{1t+2}^e + \dots + i_{1t+n-1}^e}{n}$$

The Expectations Hypothesis

Does this hypothesis explain the three observations we started with?

1. Interest rates of different maturities will move together.
 - We can see this holds from the previous equation.
 2. Yields on short-term bonds will be more volatile than yields on long-term bonds.
 - Long-term rates are averages of short-term rates, so changing one short-term rate has little effect on the long term rate.
-

The Liquidity Premium Theory

- Risk is the key to understanding the upward slope of the yield curve.
 - Bondholders face both inflation and interest-rate risk.
 - The longer the term of the bond, the greater both types of risk.
 - Computing real return from nominal return requires a forecast of *expected* future inflation.
 - *A bond's inflation risk increases with its time to maturity.*
-

The Liquidity Premium Theory

- Interest-rate risk arises from the mismatch between the investor's investment horizon and a bond's time to maturity.
 - If a bondholder plans to sell a bond prior to maturity, changes in the interest rate generate capital gains or losses.
 - The longer the term of the bond, the greater the price changes for a given change in interest rates and the larger the potential for capital losses.
- Investors require compensation for the increase in risk they take for buying longer term bonds.

The Liquidity Premium Theory

□ We can think about bond yields as having two parts:

- One that is risk free: explained by the expectations hypothesis.
- One that is a risk premium: explained by inflation and interest-rate risk.

- Together this forms the **liquidity premium theory of the term structure** of interest rates.

$$i_{nt} = rp_n + \frac{i_{1t} + i_{1t+1}^e + i_{1t+2}^e + \dots + i_{1t+n-1}^e}{n}$$



APPLYING THE CONCEPT

THE FLIGHT TO QUALITY

-
- During financial crises, people sell risky investments & buy safe ones.
 - An increase in the demand for government bonds coupled with a decrease in the demand for virtually everything else is called a **flight to quality**.
 - This leads to an increase in the risk spread.
-